Diskrete Mathematik – Woche 5

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https://github.com/tschai-yim/eth-resources
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Foundations of Set Theory

Problem 1 ⋆

Prove or disprove for all sets a, B, C. Counterexamples may only use sets built from $\varnothing$.

1. $a \in B \land B \subseteq C \Rightarrow a \in C$
2. $a \in B \land B \in C \Rightarrow a \in C$
3. $a \subseteq B \land B \in C \Rightarrow a \in C$
4. $a \in B \land B \in C \Rightarrow a \subseteq C$
5. $a \in B \Leftrightarrow \{a\} \subseteq B$

Problem 2 ⋆

Consider the formal proof of Lemma 3.6 ($\varnothing \subseteq A$) in the script.

1. Two steps are only "$\Rightarrow$", not "$\Leftrightarrow$". Why does this not invalidate the proof?
2. Give a direct proof of $\varnothing \subseteq A$ (no contradiction).
3. Prove or disprove: $\forall A(\varnothing \in A)$.
4. Prove or disprove: $\forall A(A \subseteq \varnothing \to A = \varnothing)$.

Problem 3 ⋆

Let $A = \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}\}$. No justification is required.

1. List all x with $x \in A$ and $x \subseteq A$.
2. List all subsets of A that are **not** elements of A.
3. Compute $|\{\varnothing, \{\varnothing\}, \{\varnothing, \varnothing\}, \{\{\varnothing\}, \varnothing\}, \{\varnothing, \{\varnothing\}\}\}|$.
4. List all elements of $(A \cap \{\varnothing, \{\varnothing, \varnothing\}\}) \times \{\varnothing \cup \{\varnothing\}\}$.
5. Compute $|\mathcal{P}(A \setminus \mathcal{P}(\{\varnothing\}))|$.
6. Compute $|(\varnothing \times A) \cup (\{\varnothing\} \times \{\varnothing\})|$.
7. True or false: $\mathcal{P}(\varnothing) \times \mathcal{P}(\varnothing) = \mathcal{P}(\varnothing \times \varnothing)$.

Problem 4 ⋆ ⋆

Express as formulas using only $\in$, $=$, logical operators, and quantifiers (no $\varnothing$, $\{\ldots\}$, $\cup$, $\mathcal{P}$, $|\cdot|$):

1. "A is empty"
2. "A has exactly one element"
3. "A has exactly two elements"
4. "$C = A \cup B$"
5. "$P = \mathcal{P}(A)$"

Problem 5 ⋆ ⋆

Decide whether each statement is true or false. Give a witness or counterexample, or a short proof.

1. For every set A: $A \cap \mathcal{P}(A) = \varnothing$.
2. There is a non-empty set A with $A \subseteq \mathcal{P}(A)$.
3. For all sets A, B there is a set C with $A \cup C = B$.
4. For all sets A, B there is a set C with $A \setminus C = A \cap B$.
5. There are sets A, B, C with $A \cap B$, $B \cap C$, $A \cap C$ all non-empty but $A \cap B \cap C = \varnothing$.
6. There are pairwise distinct sets A, B, C with $A \cup B = B \cup C = A \cup C$.

7. For all finite sets A, B: $|A \setminus B| = |A| - |B|$.

Operations on Sets

Problem 6 ⋆

Compute:
1. $\bigcup\{\varnothing\}$
2. $\bigcap\{\varnothing, \{\varnothing\}\}$
3. $\bigcup\{\{\varnothing\}, \{\{\varnothing\}\}\}$
4. $\bigcup\bigcup\{\{\{\varnothing\}\}\}$
5. $\bigcap\{\{\varnothing, \{\varnothing\}\}, \{\{\varnothing\}\}\}$
6. $\bigcup p$ and $\bigcap p$ for $p = \{\{a\}, \{a, b\}\}$

Why does the script not define $\bigcap\varnothing$?

Problem 7 ⋆ ⋆

Prove for all sets A, B, C:
1. $(A \setminus B) \setminus C = A \setminus (B \cup C)$
2. $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$
3. $(A \setminus B) \setminus C = A \setminus (B \setminus C) \Leftrightarrow A \cap C = \varnothing$
4. $A \setminus B = B \setminus A \Leftrightarrow A = B$
5. $A \setminus (A \setminus B) = A \cap B$. For which A, B does $A \setminus (A \setminus B) = B$ hold?

Problem 8 ⋆ ⋆

Using only $\cup$, $\cap$, $\setminus$ and parentheses, write each set as an expression in which A, B, C each appear at most once.
1. $(A \setminus B) \cup (A \setminus C)$
2. $(A \cup C) \setminus (B \cup C)$
3. $(A \cap B) \cup (A \setminus C)$
4. Prove that $(A \setminus B) \cup (B \setminus A)$ **cannot** be written with A and B each appearing at most once.

Problem 9 ⋆ ⋆

Let A, B be sets and $\mathcal{A}$ a set of sets.
1. Prove: $\bigcup\mathcal{P}(A) = A$.
2. Prove: $\mathcal{A} \subseteq \mathcal{P}(\bigcup\mathcal{A})$. Give an example where the inclusion is strict.
3. Prove: $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.
4. Prove or disprove: $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.

Problem 10 ⋆ ⋆

Call a set T *transitive* if $x \in T \Rightarrow x \subseteq T$.
1. Which of $\{\varnothing\}$, $\{\{\varnothing\}\}$, $\{\varnothing, \{\{\varnothing\}\}\}$, $\{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}\}$ are transitive?
2. Prove: T is transitive $\Leftrightarrow T \subseteq \mathcal{P}(T) \Leftrightarrow \bigcup T \subseteq T$.
3. Prove: if T is transitive, then $\mathcal{P}(T)$ is transitive.
4. Prove: if T is transitive, then $T \cup \{T\}$ is transitive.

Problem 11 ⋆ ⋆ ⋆

Let A, B, C be sets.
1. Prove: there is a set X with $A \cup X = B$ and $A \cap X = C$ if and only if $C \subseteq A \subseteq B$. Show that in this case X is unique, and give it explicitly.

For sets S, T let $S \triangle T = (S \setminus T) \cup (T \setminus S)$ (the *symmetric difference*).
2. Prove that $\triangle$ is associative: $(A \triangle B) \triangle C = A \triangle (B \triangle C)$.
3. Prove: for all A, B there is **exactly one** set X with $A \triangle X = B$.

Ordered Pairs and Cartesian Products

Problem 12 ⋆ ⋆

Prove for all sets A, B:
1. $A \times B = B \times A \Leftrightarrow A = B \vee A = \varnothing \vee B = \varnothing$
2. $A \times A = B \times B \Rightarrow A = B$

Problem 13 ⋆ ⋆

Prove or disprove. Repair false claims with an additional hypothesis.
1. $A \times B \subseteq C \times D \Rightarrow A \subseteq C$
2. $A \times B = A \times C \Rightarrow B = C$
3. $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$
4. $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$
5. Let A, B be finite with $|A| = m$, $|B| = n$, $|A \cap B| = k$. Compute $|(A \times B) \cup (B \times A)|$.

Problem 14 ⋆ ⋆

1. List $\mathcal{P}(A \times B)$ and $\mathcal{P}(A) \times \mathcal{P}(B)$ for $A = \{1\}$, $B = \{2\}$.
2. For finite A, B with $|A| = m$, $|B| = n$: for which m, n do $\mathcal{P}(A \times B)$ and $\mathcal{P}(A) \times \mathcal{P}(B)$ have the same cardinality?
3. Prove: $\mathcal{P}(A \times B) \neq \mathcal{P}(A) \times \mathcal{P}(B)$ for all sets A, B.